

ShiftLift User Guide

Student Worker Shift Management System

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1. Introduction

ShiftLift is a comprehensive web-based application designed to streamline the management of student worker schedules, shifts, and workstations. Built with Spring Boot and Vaadin, it provides an intuitive interface for both administrators and student workers.

Key Features

- **Visual Calendar:** Interactive weekly calendar with half-hour time slots
- **Automatic Conflict Detection:** Prevents double-booking of workers and workstations
- **Seniority System:** Automatic seniority calculation based on hire date
- **Max Hours Protection:** Prevents students from exceeding weekly hour limits
- **Color-Coded Workstations:** Easy visual identification of different workstations
- **PDF Export:** Download and print weekly schedules

2. Getting Started

Accessing the Application

Navigate to the application URL (typically `www.shiftlift-065i.onrender.com` in development) in your web browser.

Login

- 1 **Enter your username** in the first field

2 **Enter your password** in the second field

3 **Click "Login"** to access the system

Note: Your account type (Administrator or Student Worker) determines which features you can access.

Main Navigation

After logging in, you'll see the Pending Schedule (unpublished) calendar view. Use the hamburger menu (≡) on the left to access different features:

For Administrators:

- **View Published Schedule** - View the current published schedule
- **Manage Workers** - Add, edit, or delete student workers
- **Manage Workstations** - Add, edit, or delete workstations
- **Manage Schedules** - Create and publish schedules
- **Create New Shift** - Add shifts to the unpublished schedule
- **Download Schedule PDF** - Export current schedule as PDF
- **Download Hours Report** - Export worker hours report (from unpublished schedule)
- **Change Password** - Update your account password

For Student Workers:

- **View Published Schedule** - View the current published schedule
- **Request New Shift** - Request shifts using unpublished schedule dates (Main Menu only)
- **Download Schedule PDF** - Export current schedule as PDF
- **Change Password** - Update your account password

Understanding Pending vs. Published Schedules

Pending Schedule (Main Menu)

This is the unpublished schedule where administrators create and modify shifts for future weeks. Changes made here are not visible to student workers until published.

Published Schedule

This is the official, published schedule visible to all users. Once a schedule is published, it becomes read-only for student workers. Only administrators can edit or create shifts on published schedules.

3. Administrator Features

3.1 Managing Users

Adding a New Worker

- 1 Click **Manage Workers** in the navigation menu
- 2 Click **Add New Worker** button
- 3 Fill in the required information:
 - 4 **Username:** Unique identifier for the worker
 - 5 **Email:** Worker's email address
 - 6 **Password:** Initial password (user can change later)
 - 7 **Hire Date:** Affects seniority calculation

- **8 Max Hours:** Maximum weekly hours (default: 20)
- **9 Worker Type:** Student Worker or Manager

10 Click **Save** to create the worker account

Editing a Worker

- 1** Navigate to **Manage Workers**
- 2** Find the worker in the list
- 3** Click the **Edit** button next to their name
- 4** Modify the desired fields (password, max hours, etc.)
- 5** Click **Update** to save changes

Warning: Deleting a worker will automatically remove all their assigned shifts. Seniority numbers for remaining workers will be recalculated.

3.2 Managing Workstations

Adding a Workstation

- 1** Navigate to **Manage Workstations**
- 2** Click **Add New Workstation**
- 3** Enter the workstation name
- 4** Click **Save**

Each workstation is assigned a unique color in the calendar for easy identification.

Deleting a Workstation

Warning: Deleting a workstation will automatically remove all shifts assigned to that workstation.

3.3 Managing Schedules

Creating a New Schedule

- 1 Navigate to **Manage Schedules**
- 2 Click **Create New Schedule**
- 3 Select the **Start Date** (must be a Monday)
- 4 Select the **End Date** (must be a Friday)
- 5 Click **Save**

Important: The system allows only 1 published and 1 unpublished schedule at a time. Delete expired schedules before creating new ones.

Publishing a Schedule

- 1 Navigate to **Manage Schedules**
- 2 Select the unpublished schedule
- 3 Review all shifts to ensure accuracy
- 4 Click **Publish**

Once published, the schedule becomes visible to all student workers.

3.4 Managing Shifts

Important: Shifts can be created from both the Pending Schedule (Main Menu) and the Published Schedule views. The date constraints depend on which view you're using:

- **Main Menu (Pending Schedule):** Uses unpublished schedule dates - administrators create shifts for the upcoming schedule being prepared
- **Published Schedule View:** Uses published schedule dates - only administrators can create or edit shifts on the currently active schedule

Days outside the relevant schedule date range are greyed out and disabled.

Adding Shifts - Method 1: Calendar View

- 1 From the Main Menu (uses unpublished schedule dates) or Published Schedule view (uses published schedule dates), locate the day you want to add a shift
- 2 Verify the day is within the relevant schedule period (not greyed out)
- 3 Click on the day column in the calendar
- 4 A dialog will appear with the date pre-selected
- 5 Select the **Worker** from the dropdown
- 6 Select the **Workstation** from the dropdown
- 7 Choose **Start Time** (8:00 AM - 5:00 PM)
- 8 Choose **End Time** (must be after start time)
- 9 Click **Add Shift**

Adding Shifts - Method 2: Navigation Menu (Main Menu Only)

- 1 Click **Create New Shift** in the navigation menu (available on Main Menu - uses unpublished schedule dates)

- 2 Select the shift **Date** (must be within the unpublished schedule period)
- 3 Select the **Worker, Workstation, and Times**
- 4 Click **Add Shift**

Editing a Shift

- 1 Click on a shift block in the calendar
- 2 On the **Published Schedule view**, only administrators can edit shifts (dates constrained to published schedule)
- 3 On the **Main Menu (Pending Schedule)**, administrators can edit any shift and student workers can request shifts for themselves (dates constrained to unpublished schedule)
- 4 Modify the worker, workstation, or time fields
- 5 Click **Save** to apply changes

Restriction: On the Published Schedule view, non-admin users cannot create or edit any shifts. They can only view the schedule. Student workers can only create shifts from the Main Menu using the unpublished schedule dates.

Deleting a Shift

- 1 Click on the shift block in the calendar
- 2 Click the **Delete** button
- 3 Confirm the deletion in the dialog box

4. Student Worker Features

Viewing Schedules

Viewing the Pending Schedule

After logging in, you'll see the Pending Schedule (Main Menu). This shows the schedule being prepared for future weeks:

- Click **View Pending Schedule** from any view to return to this calendar
- Use the **arrow buttons** to navigate between weeks
- Your assigned shifts appear as colored blocks
- The **color legend** at the bottom shows which color represents each workstation

Viewing the Published Schedule

To see the official, current schedule:

- Click **View Published Schedule** in the navigation menu
- This shows the finalized schedule that is currently in effect
- Days outside the schedule period appear greyed out
- You can view but not edit shifts on this schedule

Requesting Shifts

- 1 Navigate to the **Pending Schedule** (Main Menu)
- 2 Click on a day within the unpublished schedule period (non-greyed days)
- 3 A dialog will appear - select your desired workstation and time
- 4 Click **Add Shift** to submit your request

Note: Non-admin users can only create shifts for themselves and only from the Main Menu (Pending Schedule view) using the unpublished schedule dates. You cannot create or edit shifts on the Published Schedule view.

Understanding Greyed Out Days

Days that appear greyed out with reduced opacity are outside the current schedule period:

- You cannot create shifts on these days
- The tooltip will indicate "This day is outside the schedule period"
- These typically occur when viewing weeks that extend beyond the schedule dates

Understanding Your Seniority

Your seniority is automatically calculated based on your hire date:

- **Lower number = Higher seniority**
- Senior workers can override junior workers' shifts (when workstation alternatives exist)
- You'll be notified when attempting to override another worker's shift

Changing Your Password

- 1 Click **Change Password** in the navigation menu
- 2 Enter your **current password**
- 3 Enter your **new password**
- 4 Re-enter your **new password** to confirm
- 5 Click **Change Password**

5. Seniority and Override System

How Seniority Works

Aspect	Details
Calculation	Automatically calculated based on hire date
Ranking	Lower number = higher seniority (e.g., 1 is most senior)
Updates	Recalculated when workers are added or removed
Managers	Assigned seniority number 0 (highest)

Senior Worker Override

When a senior worker requests a workstation occupied by a junior worker:

- 1 System checks if an alternative workstation is available
- 2 If available, a **confirmation dialog** appears
- 3 Dialog explains that the junior worker will be reassigned
- 4 Senior worker clicks **Override** to confirm
- 5 Junior worker is automatically moved to the available workstation
- 6 Senior worker gets the requested workstation

Limitation: Senior override only works when there's at least one other available workstation. If no alternatives exist, the override is blocked.

Admin Override

Administrators have special override privileges with two scenarios:

Scenario 1: Alternative Workstation Available

Outcome: The conflicting worker is automatically reassigned to the available workstation. Both shifts remain active.

Scenario 2: No Alternative Workstation

Warning: The admin receives a clear warning that the conflicting worker's shift will be DELETED. The admin must explicitly confirm this action.

6. Validation Rules

Shift Creation Rules

Rule	Description
✓ Time Validation	Start time must be before end time
✓ Operating Hours	Shifts must be between 8:00 AM and 5:00 PM
✓ No Double-Booking	Worker cannot have overlapping shifts
✓ Max Hours Check	Student workers cannot exceed weekly max hours (default: 20)
✓ Workstation Conflict	Each workstation can only have one shift at a time (unless overridden)

Schedule Management Rules

Rule	Description
✓ Schedule Limit	Maximum 1 published schedule at a time
✓ Unpublished Limit	Maximum 1 unpublished schedule at a time
✓ Date Range	Must start on Monday and end on Friday
✓ Date Order	End date must be after start date

User Management Rules

Rule	Description
✓ Unique Username	Each username must be unique in the system
✓ Unique Email	Each email must be unique in the system
✓ Hire Date	Determines seniority ranking
✓ Max Hours	Must be a positive number

7. Troubleshooting

Common Issues and Solutions

Problem: Cannot Create Schedule

Possible Causes:

- Already have 1 published + 1 unpublished schedule

- Selected dates don't start on Monday or end on Friday

Solution: Delete expired schedules or adjust your date selection.

Problem: Worker Double-Booked Error

Cause: The selected worker already has a shift at that time.

Solution: Choose a different time slot or different worker.

Problem: Workstation Occupied

Cause: Another worker is assigned to this workstation at the selected time.

Solutions:

- Choose a different workstation
- Choose a different time
- If you have seniority or admin privileges, use the override feature

Problem: Max Hours Exceeded

Cause: Adding this shift would cause the student worker to exceed their weekly hour limit.

Solutions:

- Reduce the shift duration
- Choose another worker
- Admin: Increase the worker's max hours if appropriate

Problem: Cannot Delete User/Workstation

Cause: Database connection issue or constraint violation.

Solution: The system automatically handles cascade deletion. If errors persist, contact technical support.

Tips for Best Results

Tip: Always review the calendar after making changes to verify shifts were created correctly.

Tip: Use the color legend at the bottom of the calendar to quickly identify which workstation each shift represents.

Tip: Download PDF schedules regularly for backup and distribution to workers without system access.

Tip: Delete old schedules regularly to keep the system clean and prevent hitting schedule limits.

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For additional support, contact your system administrator or development team.